Lecture 13 Scribe

Tutorial_1.pdf
PDF

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The relevant assumptions and conditions

Theorems, results, or standard formulas applied

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Include commentary about the prompting process

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CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) — Exam-Oriented Scribe

Based strictly on Tutorial-1 (February 13, 2026)

Tutorial_1

Q1. Partitioning 20 Dishes into 4 Platters

Given

- 20 distinct dishes.
- To be divided into 4 groups of 5 dishes each.
- Order of platters does **not** matter.

(a) Total Number of Ways to Divide into 4 Unordered Groups of 5

Definitions and Notation

- All 20 dishes are distinct.
- Each platter contains exactly 5 dishes.
- The 4 platters are indistinguishable.

Step-by-Step Derivation

1. First count the number of ways to divide 20 distinct objects into 4 *ordered* groups of 5:

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

2. Since the platters are unordered, divide by the number of permutations of 4 groups:

$$4!$$

Final Expression

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

(b) Probability All Platters Have Dishes of Same Cuisine

Given

- 5 dishes from each of 4 cuisines.
- Total dishes = 20.

Total Possible Divisions

Same as part (a).

Favorable Outcomes

Each platter contains exactly 5 dishes of a single cuisine.
Since each cuisine has exactly 5 dishes:

- Each cuisine must occupy exactly one platter.
- Number of ways to assign cuisines to platters = number of permutations of 4 cuisines:

$$4!$$

Probability

$$\frac{4!}{\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}$$

Q2. Uniform Arrival and Waiting Time

Given

- Buses every 15 minutes starting at 7 A.M.
- Passenger arrival ~ Uniform(0,30) minutes.
- Waiting time determined by distance to next bus.

Let $T \sim U(0, 30)$

(a) Waiting < 5 Minutes

Waiting time < 5 if arrival occurs:

- Between 7:10 and 7:15
- Between 7:25 and 7:30

Total favorable interval length:

$$5 + 5 = 10$$

Total interval = 30

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting > 10 Minutes

Waiting > 10 if arrival occurs:

- Between 7:00 and 7:05
- Between 7:15 and 7:20

Total favorable length:

$$5 + 5 = 10$$

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability

Given

$$P(Late) = 0.15$$

$$P(EarlyLeave) = 0.25$$

$$P(Late \cap EarlyLeave) = 0.08$$

We want:

$$P(\text{Arrives Early} \mid \text{Leaves Late})$$

Define:

- Arrives Early = complement of Late
- Leaves Late = complement of EarlyLeave

$$P(ArrivesEarly) = 1 - 0.15 = 0.85$$

$$P(LeavesLate) = 1 - 0.25 = 0.75$$

$$P(ArrivesEarly \cap LeavesLate) = 1 - P(Late \cup EarlyLeave)$$

Using inclusion-exclusion:

$$P(Late \cup EarlyLeave) = 0.15 + 0.25 - 0.08 = 0.32$$

$$P(ArrivesEarly \cap LeavesLate) = 1 - 0.32 = 0.68$$

Conditional probability:

$$\frac{0.68}{0.75}$$

Q4. Poisson Distribution (λ = 0.2)

PMF

$$P(X = k) = e^{-0.2}\frac{0.2^k}{k!}$$

(a) 0 Errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or More Errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(1) = e^{-0.2}(0.2)$$

$$P(X \geq 2) = 1 - e^{-0.2}(1 + 0.2)$$

Q5. Poisson (λ = 3.5)

$$P(X = k) = e^{-3.5}\frac{3.5^k}{k!}$$

(a) At Least 2

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$= 1 - e^{-3.5}(1 + 3.5)$$

(b) At Most 1

$$P(X \leq 1) = e^{-3.5}(1 + 3.5)$$

Q6. Poisson (λ = 3)

(a)

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-3}\left(1 + 3 + \frac{9}{2}\right)$$

(b)

$$P(X \geq 3 \mid X \geq 1)$$

$$= \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - P(0) = 1 - e^{-3}$$

Q7. Normal Distribution

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Mean = -3
Variance = 4
Standard deviation = 2

Standardization:

$$Z = \frac{X+3}{2}$$

Use Q-function:

$$Q(a) = P(Z > a)$$

---

1. $P(X \leq 0)$

$$Z = \frac{0+3}{2} = 1.5$$

$$P(X \leq 0) = 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{4+3}{2} = 3.5$$

$$P(X > 4) = Q(3.5)$$

---

3. $P(|X+3| < 2)$

$$|Z| < 1$$

$$= 1 - 2Q(1)$$

---

4. $P(|X-2| > 1)$

Transform bounds accordingly:

$$P(X > 3) + P(X < 1)$$

Express in Q-function form.

Q8. Jury Decision

Given

- 12 jurors
- Need ≥8 correct votes
- Each correct with probability θ
- Independent

Let $X \sim Binomial(12, \theta)$

$$P(\text{Correct}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9. PMF

$$p(i) = c\frac{\lambda^i}{i!}$$

Since total probability = 1:

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1$$

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - \left[e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right)\right]$$

Q10. System Reliability

Given

- n components
- Independent
- Each works with probability p
- System works if ≥ half function

Let $X \sim Binomial(n, p)$

(a) Compare n=5 and n=3

5-component system works if ≥3 function:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

3-component system works if ≥2:

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

Solve:

$$P_5 > P_3$$

(b) General Case

Compare:

$$\sum_{k=k+1}^{2k+1} \binom{2k+1}{k} p^k (1-p)^k$$

and

$$\sum_{k=k}^{2k-1} \binom{2k-1}{k} p^k (1-p)^{k-1}$$

Condition determines when larger system is better.

End of Lecture 13 Tutorial Scribe

(Strict reconstruction based on Tutorial-1)

Tutorial_1

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Lecture 13 (Tutorial) — Exam-Oriented Scribe

Based strictly on Tutorial-1 (February 13, 2026)

Tutorial_1

Q1. Partitioning 20 Distinct Dishes into 4 Platters

Given

- Total number of dishes: 20 (all distinct).
- To be divided into 4 groups (platters).
- Each platter must contain exactly 5 dishes.
- The order of platters does **not** matter.

(a) Total Number of Ways

Definitions and Assumptions

- All 20 dishes are distinct objects.
- We partition them into 4 subsets of size 5.
- Since the order of platters does not matter, the groups are unordered.

Step 1: Count Ordered Groups

If the 4 platters were distinguishable, the number of ways would be:

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

This corresponds to:

- Choosing 5 dishes for the first platter,
- 5 of the remaining 15 for the second,
- 5 of the remaining 10 for the third,
- The remaining 5 for the fourth.

Step 2: Remove Ordering of Platters

Since the 4 platters are not ordered, every unordered partition has been counted $4!$ times.

Hence the total number of valid divisions is:

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

(b) Probability All Platters Contain a Single Cuisine

Additional Condition

- There are 5 dishes from each of the four cuisines:
  - Italian
  - Indian
  - Chinese
  - Mexican

Thus total dishes = $4 \times 5 = 20$.

Total Possible Outcomes

Same as part (a):

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

Favorable Outcomes

For all platters to contain dishes of the same cuisine:

- Each cuisine has exactly 5 dishes.
- Therefore each cuisine must occupy exactly one platter.
- The only freedom is assigning cuisines to platters.

Number of ways:

$$4!$$

Required Probability

$$\frac{4!}{\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}$$

Q2. Waiting Time with Uniform Arrival

Given

- Buses arrive every 15 minutes starting at 7:00 A.M.

- Passenger arrival time is uniformly distributed between 7:00 and 7:30.

Let the arrival time T be uniformly distributed on $[0, 30]$.

(a) Waiting Time < 5 Minutes

Waiting time is the time until the next bus.

Bus arrival times:

- 0 minutes (7:00)
- 15 minutes
- 30 minutes

Passenger waits less than 5 minutes if arrival occurs:

- Between 10 and 15 minutes,
- Between 25 and 30 minutes.

Total favorable length:

$$5 + 5 = 10$$

Total interval length:

$$30$$

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting Time > 10 Minutes

Waiting exceeds 10 minutes if arrival occurs:

- Between 0 and 5 minutes,
- Between 15 and 20 minutes.

Total favorable length:

$$5 + 5 = 10$$

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability

Given

$$P(L) = 0.15$$

$$P(E) = 0.25$$

$$P(L \cap E) = 0.08$$

Where:

- L: arrives late
- E: leaves early

We need:

$$P(\text{Arrives early} \mid \text{Leaves late})$$

Step 1: Complements

Arrives early = L^c
Leaves late = E^c

$$P(L^c) = 1 - 0.15 = 0.85$$

$$P(E^c) = 1 - 0.25 = 0.75$$

Step 2: Compute $P(L \cup E)$

Using inclusion–exclusion:

$$P(L \cup E) = P(L) + P(E) - P(L \cap E) = 0.15 + 0.25 - 0.08 = 0.32$$

Step 3: Complement

$$P(L^c \cap E^c) = 1 - P(L \cup E) = 1 - 0.32 = 0.68$$

Step 4: Conditional Probability

$$P(L^c \mid E^c) = \frac{P(L^c \cap E^c)}{P(E^c)} = \frac{0.68}{0.75}$$

Q4. Poisson with Mean 0.2

Given

Expected number of errors per page = 0.2.

Thus:

$$X \sim \text{Poisson}(0.2)$$

PMF:

$$P(X = k) = e^{-0.2} \frac{0.2^k}{k!}$$

(a) 0 Errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or More Errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(1) = e^{-0.2}(0.2)$$

$$P(X \geq 2) = 1 - e^{-0.2}(1 + 0.2)$$

Q5. Poisson with Mean 3.5

$$X \sim \text{Poisson}(3.5)$$

(a) At Least 2

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-3.5}(1 + 3.5)$$

(b) At Most 1

$$P(X \leq 1) = e^{-3.5}(1 + 3.5)$$

Q6. Poisson with λ = 3

$$X \sim \text{Poisson}(3)$$

(a)

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-3}\left(1 + 3 + \frac{9}{2}\right)$$

(b)

$$P(X \geq 3 \mid X \geq 1) = \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - e^{-3}$$

Q7. Continuous Random Variable

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

This corresponds to:

Mean $= -3$
Variance $= 4$
Standard deviation $= 2$

Standardization:

$$Z = \frac{X+3}{2}$$

Use Q-function notation.

---

1. $P(X \leq 0)$

$$Z = \frac{0+3}{2} = 1.5$$

$$P(X \leq 0) = 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{4+3}{2} = 3.5$$

$$P(X > 4) = Q(3.5)$$

---

3. $P(|X+3| < 2)$

$$|Z| < 1$$

$$= 1 - 2Q(1)$$

---

4. $P(|X-2|>1)$

$$P(X>3)+P(X<1)$$

Express in Q-function form using standardization.

Q8. Jury Decision

Given

- 12 jurors.
- Conviction requires ≥ 8 guilty votes.
- Each juror independently makes correct decision with probability θ.

Let:

$$X \sim \text{Binomial}(12, \theta)$$

Correct decision occurs if:

$$X \geq 8$$

Thus:

$$P(\text{Correct}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9. PMF with Unknown Constant

$$p(i) = c\frac{\lambda^i}{i!}$$

Step 1: Determine c

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1$$

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X=0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right)$$

Q10. System Reliability

Given

- n independent components.
- Each functions with probability p.
- System works if at least half function.

Let:

$$X \sim \text{Binomial}(n, p)$$

(a) Compare 5 vs 3 Components

5-component works if $X \geq 3$:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

3-component works if $X \geq 2$:

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

We compare $P_5 > P_3$.

(b) General Case

Compare:

$$\sum_{k=k+1}^{2k+1} \binom{2k+1}{k} p^k (1-p)^k$$

with

$$\sum_{k=k}^{2k-1} \binom{2k-1}{k} p^k (1-p)^{k-1}$$

This determines when the larger system is more reliable.

End of Lecture 13 Tutorial Scribe

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Lecture 13 (Tutorial) — Exam-Oriented Scribe

Based strictly on Tutorial-1 (February 13, 2026)

Tutorial_1

Q1. Division of 20 Distinct Dishes into 4 Platters

Problem Setting

- 20 distinct dishes.
- To be divided into 4 platters.
- Each platter must contain exactly 5 dishes.
- Order of platters does **not** matter.

(a) Total Number of Ways

Definitions and Assumptions

- All dishes are distinct.
- We form 4 subsets, each of size 5.
- The 4 subsets are unordered.

Step 1: Count Ordered Groups

Choose 5 dishes for the first group:

$$\binom{20}{5}$$

Choose 5 of the remaining 15:

$$\binom{15}{5}$$

Choose 5 of the remaining 10:

$$\binom{10}{5}$$

Choose 5 of the remaining 5:

$$\binom{5}{5}$$

Total (ordered groups):

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

Step 2: Remove Ordering

Since the order of platters does not matter, divide by $4!$:

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

(b) Probability Each Platter Contains One Cuisine

Additional Information

- 5 dishes from each cuisine.
- Total cuisines = 4.
- Total dishes = 20.

Total Possible Divisions

Same as part (a):

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

Favorable Outcomes

Since each cuisine has exactly 5 dishes:

- Each cuisine must form exactly one platter.
- Only assignment of cuisines to platters varies.

Number of assignments:

$$4!$$

Required Probability

$$\frac{4!}{\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}$$

Q2. Waiting Time with Uniform Arrival

Problem Setting

- Buses arrive every 15 minutes starting at 7:00 A.M.
- Arrival time uniformly distributed between 7:00 and 7:30.

Let $T \sim U(0, 30)$.

(a) Waiting Less Than 5 Minutes

Waiting time < 5 if arrival occurs:

- Between 10 and 15 minutes,
- Between 25 and 30 minutes.

Total favorable length:

$$5 + 5 = 10$$

Total interval length:

$$30$$

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting More Than 10 Minutes

Waiting > 10 if arrival occurs:

- Between 0 and 5 minutes,
- Between 15 and 20 minutes.

Total favorable length:

$$5 + 5 = 10$$

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability

Given

$$P(L) = 0.15$$

$$P(E) = 0.25$$

$$P(L \cap E) = 0.08$$

Where:

- L: arrives late
- E: leaves early

We seek:

$$P(L^c \mid E^c)$$

Step 1: Complements

$$P(L^c) = 1 - 0.15 = 0.85$$

$$P(E^c) = 1 - 0.25 = 0.75$$

Step 2: Inclusion–Exclusion

$$P(L \cup E) = P(L) + P(E) - P(L \cap E) = 0.15 + 0.25 - 0.08 = 0.32$$

Step 3: Complement

$$P(L^c \cap E^c) = 1 - P(L \cup E) = 1 - 0.32 = 0.68$$

Step 4: Conditional Probability

$$P(L^c \mid E^c) = \frac{P(L^c \cap E^c)}{P(E^c)} = \frac{0.68}{0.75}$$

Q4. Poisson Distribution (Mean = 0.2)

Assumption

Expected number of errors per page = 0.2.

Thus:

$$X \sim \text{Poisson}(0.2)$$

PMF:

$$P(X = k) = e^{-0.2}\frac{0.2^k}{k!}$$

(a) 0 Errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or More Errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(1) = e^{-0.2}(0.2)$$

$$P(X \geq 2) = 1 - e^{-0.2}(1 + 0.2)$$

Q5. Poisson Distribution (Mean = 3.5)

$$X \sim \text{Poisson}(3.5)$$

(a) At Least 2

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-3.5}(1 + 3.5)$$

(b) At Most 1

$$P(X \leq 1) = e^{-3.5}(1 + 3.5)$$

Q6. Poisson (λ = 3)

$$X \sim \text{Poisson}(3)$$

(a) $P(X \geq 3)$

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-3}\left(1 + 3 + \frac{9}{2}\right)$$

(b) $P(X \geq 3 \mid X \geq 1)$

$$P(X \geq 3 \mid X \geq 1) = \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - e^{-3}$$

Q7. Continuous Random Variable

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Mean = -3
Variance = 4
Standard deviation = 2

Standardization:

$$Z = \frac{X+3}{2}$$

---

1. $P(X \leq 0)$

$$Z = \frac{0+3}{2} = 1.5$$

$$P(X \leq 0) = 1 - Q(1.5)$$

---

2. $P(X > 4)$

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$$P(X > 4) = Q(3.5)$$

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3. $P(|X+3| < 2)$

$$|Z| < 1$$

$$= 1 - 2Q(1)$$

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4. $P(|X-2| > 1)$

$$P(X > 3) + P(X < 1)$$

Express using Q-function after standardization.

Q8. Jury Decision

Given

- 12 jurors.
- At least 8 must vote guilty.
- Each juror independently correct with probability θ.

Let:

$$X \sim \text{Binomial}(12, \theta)$$

Correct decision if:

$$X \geq 8$$

$$P(\text{Correct}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9. PMF $p(i) = c\frac{\lambda^i}{i!}$

Step 1: Determine Constant

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1$$

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

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Given

- n independent components.
- Each functions with probability p.
- System works if at least half function.

Let:

$$X \sim \text{Binomial}(n, p)$$

(a) Compare 5 and 3 Components

5-component system works if $X \geq 3$:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

3-component system works if $X \geq 2$:

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

Compare:

$$P_5 > P_3$$

(b) General Case

Compare:

$$\sum_{k=k+1}^{2k+1} \binom{2k+1}{k} p^k (1-p)^k$$

with

$$\sum_{k=k}^{2k-1} \binom{2k-1}{k} p^k (1-p)^{k-1}$$

This determines when the $(2k+1)$-component system is better than the $(2k-1)$-component system.

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The underlying definitions and notation used

The relevant assumptions and conditions

Theorems, results, or standard formulas applied

Step-by-step derivations or proofs when present in the context

Clear and logically structured worked solutions (only if they appear in the provided material)

While writing the scribe, reason step by step through all definitions, assumptions, derivations, and example solutions exactly as presented in the context. Make the logical flow explicit so that a student can reconstruct the reasoning during revision.

The structure of the scribe must:

Follow the order of topics/questions as presented in Lecture 13

Use clear section headings and mathematical notation

Maintain a formal academic tone

Preserve logical dependencies between concepts

Do not:

Add new examples

Simplify beyond what appears in the lecture

Introduce alternative solution methods

Include commentary about the prompting process

Omit intermediate reasoning steps that are shown in the lecture

The final output must function as a faithful, exam-ready reconstruction of Lecture 13.

CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) — Exam-Oriented Scribe

Based strictly on Tutorial-1 (February 13, 2026)
Tutorial_1

Q1. Division of 20 Distinct Dishes into Four Platters

Problem Data

- Total dishes: 20 (all distinct).
- To be divided into 4 platters.
- Each platter must contain exactly 5 dishes.
- The order of platters does not matter.

(a) Total Number of Ways

Definitions and Assumptions

- We partition 20 distinct objects into 4 subsets of equal size (5 each).
- Subsets are unordered.

Step 1: Count Ordered Selections

Choose 5 dishes for the first platter:

$$\binom{20}{5}$$

Choose 5 of the remaining 15:

$$\binom{15}{5}$$

Choose 5 of the remaining 10:

$$\binom{10}{5}$$

Choose the remaining 5:

$$\binom{5}{5}$$

Thus, total ordered groupings:

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

Step 2: Remove Ordering of Platters

Since platters are indistinguishable, divide by $4!$:

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

(b) Probability All Platters Contain One Cuisine Each

Additional Information

- 5 dishes from each of four cuisines.
- Total dishes = 20.

Total Possible Divisions

Same as in part (a):

$$\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}$$

Favorable Divisions

- Each cuisine has exactly 5 dishes.
- Hence each cuisine must form one platter.
- Only the assignment of cuisines to platters varies.

Number of assignments:

$$4!$$

Probability

$$\frac{4!}{\frac{1}{4!}\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}$$

Q2. Waiting Time with Uniform Arrival

Problem Data

- Buses arrive every 15 minutes starting at 7:00 A.M.
- Passenger arrival time uniformly distributed between 7:00 and 7:30.

Let arrival time $T \sim U(0, 30)$.

(a) Waiting Less Than 5 Minutes

Waiting time is the time until the next bus.

Bus arrival times:

- 0 minutes
- 15 minutes
- 30 minutes

Arrival results in waiting < 5 minutes if:

- Arrival between 10 and 15,
- Arrival between 25 and 30.

Total favorable length:

$$5 + 5 = 10$$

Total interval length:

$$30$$

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting More Than 10 Minutes

Waiting > 10 minutes if:

- Arrival between 0 and 5,
- Arrival between 15 and 20.

Total favorable length:

$$5 + 5 = 10$$

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability

Given

$$P(L) = 0.15$$

$$P(E) = 0.25$$

$$P(L \cap E) = 0.08$$

Where:

- L: arrives late,
- E: leaves early.

We seek:

$$P(L^c \mid E^c)$$

Step 1: Complements

$$P(L^c) = 1 - 0.15 = 0.85$$

$$P(E^c) = 1 - 0.25 = 0.75$$

Step 2: Inclusion–Exclusion

$$P(L \cup E) = P(L) + P(E) - P(L \cap E) = 0.15 + 0.25 - 0.08 = 0.32$$

Step 3: Complement Event

$$P(L^c \cap E^c) = 1 - P(L \cup E) = 1 - 0.32 = 0.68$$

Step 4: Conditional Probability

$$P(L^c \mid E^c) = \frac{P(L^c \cap E^c)}{P(E^c)} = \frac{0.68}{0.75}$$

Q4. Poisson Distribution (Mean 0.2)

Assumption

Expected number of typographical errors per page = 0.2.

$$X \sim \text{Poisson}(0.2)$$

PMF:

$$P(X = k) = e^{-0.2} \frac{0.2^k}{k!}$$

(a) Zero Errors

$$P(X = 0) = e^{-0.2}$$

(b) Two or More Errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(1) = e^{-0.2}(0.2)$$

$$P(X \geq 2) = 1 - e^{-0.2}(1 + 0.2)$$

Q5. Poisson Distribution (Mean 3.5)

$$X \sim \text{Poisson}(3.5)$$

(a) At Least 2 Accidents

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-3.5}(1 + 3.5)$$

(b) At Most 1 Accident

$$P(X \leq 1) = e^{-3.5}(1 + 3.5)$$

Q6. Poisson (λ = 3)

$$X \sim \text{Poisson}(3)$$

(a) $P(X \geq 3)$

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-3}\left(1 + 3 + \frac{9}{2}\right)$$

(b) Conditional Probability

$$P(X \geq 3 \mid X \geq 1) = \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - e^{-3}$$

Q7. Continuous Random Variable

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Mean = -3
Variance = 4
Standard deviation = 2

Standardization:

$$Z = \frac{X+3}{2}$$

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1. $P(X \leq 0)$

$$Z = \frac{0+3}{2} = 1.5$$

$$P(X \leq 0) = 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{4+3}{2} = 3.5$$

$$P(X > 4) = Q(3.5)$$

---

3. $P(|X+3| < 2)$

$$|Z| < 1$$

$$= 1 - 2Q(1)$$

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4. $P(|X-2| > 1)$

$$P(X > 3) + P(X < 1)$$

Express in Q-function form via standardization.

Q8. Jury Decision

Given

- 12 jurors.
- At least 8 must vote guilty.
- Each juror independently makes the correct decision with probability θ.

Let:

$$X \sim \text{Binomial}(12, \theta)$$

Correct decision if:

$$X \geq 8$$

$$P(\text{Correct}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9. PMF $p(i) = c\frac{\lambda^i}{i!}$

Determination of c

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1$$

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right)$$

Q10. System Reliability

Given

- n independent components.
- Each functions with probability p.
- System operates if at least half function.

Let:

$$X \sim \text{Binomial}(n, p)$$

(a) 5 vs 3 Components

5-component works if $X \geq 3$:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

3-component works if $X \geq 2$:

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

Compare:

$$P_5 > P_3$$

(b) General Case

Compare:

$$\sum_{k=k+1}^{2k+1} \binom{2k+1}{k} p^k (1-p)^k$$

with

$$\sum_{k=k}^{2k-1} \binom{2k-1}{k} p^k (1-p)^{k-1}$$

This determines when the $(2k+1)$-component system is better than the $(2k-1)$-component system.

End of Lecture 13 Tutorial Scribe

Tutorial_1

Sources